

Battery Swapping Station

Modular, AUTOSAR-Compliant EV Infrastructure · Built From Scratch

A ground-up modular battery swapping station that lets EV riders exchange a depleted battery for a fully charged one in minutes — eliminating charging downtime, the single biggest barrier to commercial EV adoption — delivered as 6 AUTOSAR-compliant pilot units for client field testing.

Minutes

Swap vs. hours charging

6 units

Pilot units delivered

AUTOSAR

Automotive-grade

New line

Infra revenue stream

01. WHY WE NEED THIS PROJECT

Charging time is the defining bottleneck for commercial EV fleets. A delivery rider or e-rickshaw driver loses income for every hour a vehicle sits charging. Battery swapping solves this directly, but the infrastructure to do it safely and reliably did not exist off the shelf for the client's needs.

- ▶ **Downtime kills fleet economics:** Commercial drivers are paid by trips completed; hours spent charging are lost revenue for both driver and fleet operator.
- ▶ **No reliable modular station existed:** The client needed a station built from the ground up — safe, automated, and standards-compliant — not a lab prototype.
- ▶ **Safety is non-negotiable:** Swapping involves high-energy batteries and physical actuators; any unsafe state (battery ejecting mid-charge, door unlocking wrongly) is a serious hazard and a deal-breaker for deployment.

Core thesis: Whoever builds safe, reliable swapping infrastructure captures the recurring 'energy-as-a-service' layer of the EV economy — far stickier and higher-value than one-time hardware sales.

02. BUSINESS & REVENUE IMPACT

Revenue model unlocked

- ▶ **Energy-as-a-service:** Riders pay per swap — a recurring revenue stream rather than a one-time sale, with predictable, high-frequency usage.
- ▶ **Hardware + recurring:** The station is sold/leased to operators, then generates ongoing per-swap revenue and service contracts.
- ▶ **Battery utilisation:** Centralised charging optimises battery health and grid usage, improving margin per battery over its lifetime.

Quantified value drivers

Impact Area	Mechanism	Business Outcome
Driver productivity	Swap in minutes vs. hours	More trips/day → higher driver & operator income
Recurring revenue	Per-swap pricing	Predictable, high-frequency cash flow
Battery lifetime	Optimised central charging	Lower degradation → better margin per asset

Impact Area	Mechanism	Business Outcome
Market position	First-mover on safe infra	Captures the energy-as-a-service layer

03. CUSTOMER SATISFACTION IMPACT

- ▶ **Riders / drivers:** Near-zero downtime means more earning time and no 'range anxiety' — the single biggest satisfaction driver in commercial EV use.
- ▶ **Fleet operators:** Predictable uptime across the fleet, lower battery capex risk, and a safer operation.
- ▶ **Client (station operator):** A deployable, standards-compliant product they can confidently roll out to their own customers.

Satisfaction signal: *Swapping converts the worst part of the EV experience (waiting to charge) into a 2-minute pit stop — a step-change in daily usability.*

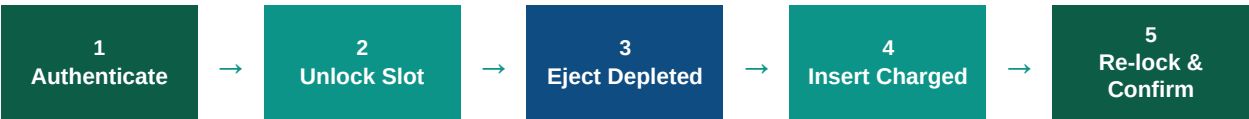
04. STAKEHOLDER ALIGNMENT

As New Product Development lead, I aligned mechanical, electrical, embedded, and client stakeholders around a single safe, shippable MVP.

Stakeholder	Their Priority	How I Aligned Them
Mechanical team	Robust physical swap mechanism	Defined clear module interfaces up front
Electrical / Embedded	Safe actuator & SoC control	Specified fail-safe states & interlocks as hard requirements
Client	Automotive-grade reliability	Committed to AUTOSAR compliance for core safety components
Field-test / Ops	Units that survive real use	Phased to 6 pilot units for controlled field testing
Leadership	On-time MVP delivery	Prioritised essential AUTOSAR scope; deferred non-critical polish

05. HOW IT WORKS — SWAP SEQUENCE

The station runs a strict, safety-gated state machine. Each transition has a defined fail-safe so an unsafe state is physically impossible, not just discouraged.



Step detail

- ▶ **1 · Authenticate:** Rider/battery is verified; the system selects an appropriate fully-charged slot based on SoC.
- ▶ **2 · Unlock Slot:** Door-lock actuator releases only the selected slot, with interlocks preventing any other slot from opening.
- ▶ **3 · Eject Depleted:** Depleted battery is safely ejected; system confirms removal before proceeding.
- ▶ **4 · Insert Charged:** Charged battery is dispensed/seated; SoC and seating are validated.
- ▶ **5 · Re-lock & Confirm:** Slot re-locks, UI confirms success and displays new SoC; depleted battery begins charging centrally.

Safety architecture

- ▶ Every actuator has an explicit fail-safe default state.
- ▶ Interlock conditions make unsafe transitions (e.g. ejecting a charging battery, unlocking mid-cycle) physically impossible.
- ▶ Modular subsystems (sequencing, actuator control, SoC display, interlocks) were built and tested in parallel against defined interfaces.

06. OUTCOME & STRATEGIC VALUE

- ▶ Delivered 6 AUTOSAR-compliant pilot units from a ground-up design with no off-the-shelf reference.

- ▶ Proved the technical and safety feasibility of modular swapping for the client's field conditions.
- ▶ Established the foundation for an energy-as-a-service revenue model with recurring per-swap economics.
- ▶ Modular architecture reduced the cost of adding capability later (e.g. a new battery type integrated in days, not weeks).

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